

SAM McDEVITT

Nashville, TN | 601-668-6782 | sam@mcdevitt.page | mcdevitt.page | linkedin.com/in/sammcdevitt | github.com/sammcdsam

SUMMARY

Computer vision engineer with 4+ years building and shipping production, real-time CV and ML inference systems — YOLO-based detection and tracking, multi-camera video analytics, and vision-language systems — on a foundation of safety-critical embedded engineering. Own the full pipeline: RTSP/GStreamer video ingestion through YOLOv8 inference, multi-object tracking, and event generation, optimized for real-time edge deployment with TensorRT/ONNX and INT8 quantization on NVIDIA Jetson and GPU hardware. Repeated record of shipping high-accuracy CV in regulated, real-world environments (DHS/TSA, DoD, healthcare).

Clearances: U.S. Secret (previously held — Boeing/Dynetics) | DHS (previously held — CBP contract); eligible for reinstatement

EXPERIENCE

AI Metrics — Lead AI/ML Engineer

June 2025 – Present

- Led modernization of the company's core AI codebase — selected the target framework, designed the migration strategy, and improved segmentation Dice scores and regression R^2 over the legacy baseline.
- Cut required radiologist corrections by 20% through improvements to a production classification model, directly reducing clinical review burden.
- Architected AWS infrastructure for distributed hyperparameter tuning of segmentation and localization models on CT/MRI imaging data in a regulated healthcare environment.
- Built production preprocessing pipelines — normalization, augmentation, ROI extraction — across multi-modal medical datasets.
- Translated clinical and regulatory requirements into model acceptance criteria with physician and business stakeholders.

Analytical AI — Senior AI/ML Engineer

May 2023 – May 2025

- Built real-time computer-vision pipelines with YOLO / YOLOv8 for object detection and multi-object tracking — foreign-object-debris (FOD) detection, and a TSA passenger-throughput prototype that ingested live RTSP feeds from multiple IP cameras to track people across zones and measure dwell time at kiosks.
- Engineered high-throughput RTSP video-ingestion pipelines (GStreamer, OpenCV) feeding YOLOv8 inference across multiple concurrent camera streams for real-time analytics.
- Built a searchable-video capability: YOLO localizes objects of interest (people, vehicles) per frame and a vision-language model generates natural-language descriptions, letting operators query recorded footage by description.
- Optimized detection models for real-time edge inference with TensorRT and ONNX Runtime — including INT8 quantization and calibration — and deployed on NVIDIA Jetson.
- Designed and deployed a production computer-vision system (detection + model ensembling) for U.S. Customs and Border Protection that scored 98.9% on the government's technical challenge — the direct basis for a \$25M, 5-year production contract award.
- Led prototype development for two U.S. Air Force SBIR contracts (\$2.5M combined); both demos secured follow-on production deployments.
- Built FastAPI services deployed on Linux via systemd for model serving and structured edge-to-cloud event delivery, with performance monitoring for production reliability.
- Engineered a real-time sensor-fusion system integrating Intel depth cameras with radiation detectors for source-carrier attribution and tracking; owned model versioning, monitoring, and CI/CD for production ML.

Boeing — Software Engineer, Advanced Development & Prototype Systems

Aug 2022 – May 2023

- Developed and verified mission-critical embedded algorithms for PAC-3 (Patriot) missile seeker systems in Ada, VHDL, and MATLAB.
- Owned requirements engineering, verification testing, and documentation in a safety-critical, highly regulated defense environment.

Dynetics — Computer Engineer III

Aug 2021 – Aug 2022

- Developed and tested VHDL for satellite communications systems; contributed to a hypersonic missile software program.
- Built a Kalman-filter-based control system in Python for a seeker testbed.

- Built CNN segmentation models in TensorFlow/Keras for autonomous-driving perception.
- Developed multi-modal sensor fusion (camera, LiDAR, depth) for ROS-based autonomous vehicle demonstrations.

PROJECTS

Secure Medical RAG Pipeline

[rag.mcdevitt.page](#) (pw: tibia-vector-7) · [github.com/sammcdsam/medical-rag-pipeline](#)

Agentic, access-controlled, federated RAG over live PubMed — air-gap-capable, exposed over MCP · Python

- Built an agentic RAG loop (Claude) over 27K live PubMed abstracts (34K chunks) that decomposes multi-hop questions and reformulates its own queries — with the caller's clearance bound so the agent cannot escalate its own access, making retrieval resistant to prompt injection from retrieved text.
- Enforced need-to-know access control as a vector-store pre-filter (unauthorized documents never reach the model) with an append-only audit log; federated retrieval across governed silos with two-level authorization and provenance-tagged, distance-merged results.
- Exposed the access-controlled retriever as an MCP server callable from any MCP client with caller clearance still enforced; pluggable generation layer (Claude API or fully local Ollama, zero code change) for air-gapped deployment.
- Hand-built chunking, retrieval, and citation with no LLM framework, over local-GPU bge-small embeddings and ChromaDB; label-free eval harness (Precision@k, nDCG@k, LLM-judge faithfulness) used to reject a reranker that failed to beat the bi-encoder baseline, plus a pytest suite verifying access controls hold on the real index.

College Baseball Predictor

[baseball.mcdevitt.page](#) · [github.com/sammcdsam/college-baseball-predictor](#)

NCAA Division I win-probability and analytics platform, live in production all season · Python

- Stacked 12 heterogeneous models (Elo, Pythagorean, Poisson, LightGBM, XGBoost, neural, situational) with an XGBoost meta-ensemble over a full season — 8,449 graded games across 313 Division I teams, yielding the best-calibrated model in the system at 0.019 ECE over 6,558 out-of-sample games.
- Engineered leak-safe ML: diagnosed and removed current-state SQL joins leaking end-of-season ratings into historical training rows (worth ~13 points of illusory accuracy), then locked the fix behind provenance-tagged predictions, a pregame timestamp cutoff, a regression test, and strict chronological walk-forward retraining.
- Automated live-data ingestion from StatBroadcast, SIDEARM, and ESPN (runners, counts, live win probability) on scheduled pipelines, feeding a self-hosted Flask dashboard served through a Cloudflare Tunnel; closing-line-value tracking benchmarks every prediction against the market.

EDUCATION

Mississippi State University — M.S., Electrical & Computer Engineering

2020 – 2022 | GPA 3.80

Mississippi State University — B.S., Computer Engineering

2016 – 2020 | GPA 3.83

PUBLICATIONS

McDevitt S, et al. "Wearables for Biomechanical Performance Optimization and Risk Assessment in Industrial and Sports Applications." *Bioengineering* (Basel), 2022.

TECHNICAL SKILLS

Computer Vision & Edge AI: YOLO / YOLOv8 (detection + tracking), multi-object & multi-camera tracking, vision-language models (VLMs), image segmentation, anomaly detection, sensor fusion, explainable AI (Grad-CAM, RISE)

Real-Time Inference & Optimization: TensorRT, ONNX Runtime, INT8 quantization & calibration, NVIDIA Jetson, GPU & edge inference, multi-stream throughput optimization

Video & Streaming: RTSP ingestion, GStreamer, OpenCV, live IP-camera pipelines, hardware-accelerated decode

Frameworks, Cloud & MLOps: PyTorch, TensorFlow/Keras, FastAPI, Docker, systemd, AWS (EC2, S3, SageMaker), CI/CD, model versioning & monitoring

GenAI & RAG: LLMs, retrieval-augmented generation, agentic systems, MCP, embeddings & vector search (ChromaDB), LLM evaluation (Precision@k, nDCG, LLM-as-judge), Claude API, Ollama

Languages & Data: Python, C++, SQL, C, MATLAB; NumPy, Pandas, scikit-learn, ETL & data-pipeline design